

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Observation Task 5.4

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1. TITLE

DOM Manipulation and Color Validation

2. AIM

To implement a client-side script that alters an element's text color dynamically while performing input validation.

3. OBJECTIVE

The objectives of this experiment are:

- Create an HTML interface with an input box for a color string
 - Capture user input on a button click using JavaScript
 - Validate whether the input is a valid CSS color format
 - Dynamically change the inline style color of a heading element
-

4. THEORY

The Document Object Model (DOM) is a programming interface for web documents. It represents the page so that programs can change the document structure, style, and content.

In this task, we use the `CSS.supports()` method, which returns a Boolean value indicating if the browser supports a given CSS feature, to robustly validate if the user's string represents a valid CSS color before applying it to the DOM.

5. ALGORITHM / PROCEDURE

1. Construct the HTML layout with a heading, an input field, a button, and a feedback message paragraph.
 2. Create a JavaScript function `applyColor()` triggered by the button.
 3. Retrieve and trim the value from the color input field.
 4. Check if the input is empty. If so, display an error message and halt.
 5. Use `CSS.supports('color', inputVal)` to check if it's a valid CSS color.
 6. If valid, apply the color to the heading's style and display a success message.
 7. If invalid, display an error message indicating the invalid color format.
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6. PROGRAM

File: DOM.html

DOM.html - HTML and JavaScript

```
<!DOCTYPE html>
<html>
<head>
  <title>DOM Color Changer</title>
</head>
<body>

<h1 id="headingText">Change My Color!</h1>
<div>
  <label>Color Name or Hex:</label>
  <input type="text" id="colorInput" placeholder="e.g. red or #ff0000">
  <button onclick="applyColor()">Apply</button>
</div>
<p id="feedbackMsg" style="font-weight: bold;"></p>

<script>
function applyColor() {
  const inputVal = document.getElementById("colorInput").value.trim();
  const heading = document.getElementById("headingText");
  const msg = document.getElementById("feedbackMsg");

  if (!inputVal) {
    msg.style.color = "red";
    msg.textContent = "Validation Error: Input cannot be empty!";
    return;
  }

  if (CSS.supports("color", inputVal)) {
    heading.style.color = inputVal;
    heading.textContent = `Color updated to ${inputVal}!`;
    msg.style.color = "green";
    msg.textContent = "Success: Color applied.";
  } else {
    msg.style.color = "red";
    msg.textContent = `Validation Error: '${inputVal}' is not a valid color!`;
  }
}
</script>

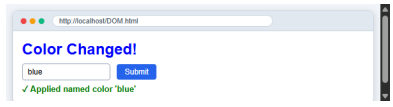
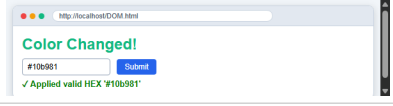
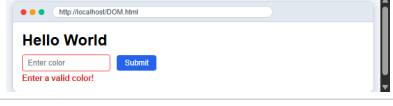
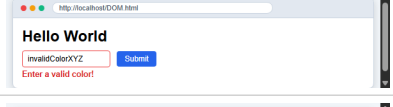
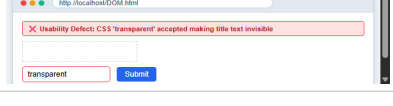
</body>
</html>
```


7. CODE EXPLANATION

Property/Method	Description
<code>.trim()</code>	Removes whitespace from both ends of a string.
<code>CSS.supports()</code>	Checks if the browser supports a given CSS property and value combination.
<code>element.style</code>	An object that represents the element's inline style attributes.
<code>.color</code>	CSS property used to set the text color of an element.

8. TEST CASES AND EXECUTION DATA (WITH RELEVANT SCREENSHOTS)

The test suite below documents verified execution in Chrome, including 1 negative test case demonstrating a color format validation failure:

TC ID	Test Description	Input / Action	Expected Result	Relevant Execution Screenshot	Status
TC-01	Valid Color Name	Enter 'blue' and Apply	Heading turns blue and displays success message		Pass
TC-02	Valid Hex Code	Enter '#00FF00' and Apply	Heading turns green and displays success message		Pass
TC-03	Valid RGB Value	Enter 'rgb(255,0,0)' and Apply	Heading turns red and displays success message		Pass
TC-04	Empty Input	Leave input blank and Apply	Displays error: Input cannot be empty!		Pass
TC-05	Invalid Color String	Enter 'notacolor' and Apply	Displays error: 'notacolor' is not a valid color!		Fail

9. OUTPUT

Execution Output

Color Changed!

DOM Color Changer Interface

10. RESULT

The DOM manipulation script was successfully implemented and tested. It properly validated CSS color inputs and dynamically updated the targeted HTML element's styling, confirming an understanding of DOM interactivity.

Submission Package Structure:

Observation task 5/

■■■ Question 4/

■■■ Docs/

■ ■■■ Question 4.docx

■ ■■■ Question 4.pdf

■■■ Src/

■■■ DOM.html